

Hall Effect Thrusters

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What is a Hall Effect Thruster?

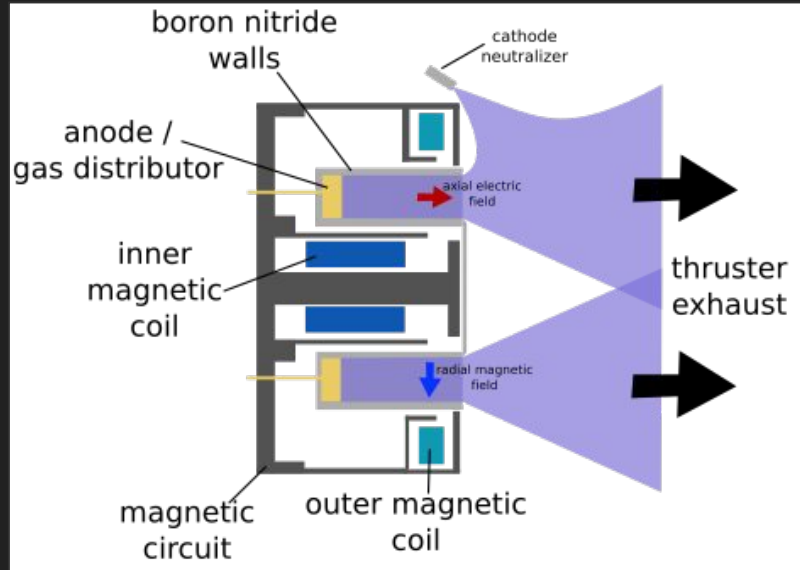


Figure 1: Diagram of Hall Thruster

Trade Offs:

Advantages	Disadvantages
High specific impulse → efficient fuel usage	Very low thrust → maneuvers take longer
High ionization efficiency → no Child-Langmuir limit	Channel wall erosion limits operational lifetime
Long operational lifetime → no grids to wear	Large plume divergence reduces efficiency
Simpler design vs. gridded ion thrusters	Complex plasma behavior → difficult to model

Hall thrusters use crossed electric and magnetic fields to ionize and accelerate propellant ions, trading raw thrust for exceptional fuel efficiency. This enables satellites to carry more payload and extend mission life.

SpaceX Starlink Thruster Overview

GEN 1: 2019-2023

Starlink V2 (1st Generation)

~4,000 satellites deployed – Falcon 9
Krypton Hall Thruster

GEN 2: 2023-Present

Starlink V2 Mini

5,413 satellites launched – Launches Ongoing
Argon Hall Thruster

V2 Mini vs. Gen 1 Improvement

2.4x

Thrust

1.5x

Isp

4x

User

Capacity

V2 Mini – Key Specs

- Propellant: Argon (Ar)
- Thrust: 170 mN
(↑+100 mN)
- Specific Impulse (Isp):
2,500 s
- Total Efficiency: 50%
- Power Draw: 4.2 kW
- Thruster Mass: 2.1 kg

Starlink V1.5



Starlink V2 Mini

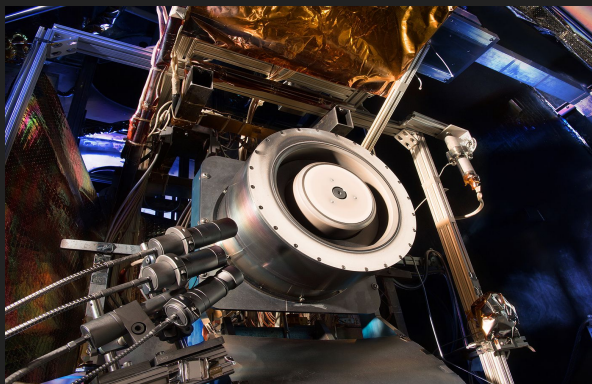


SpaceX's switch from krypton to argon Hall thrusters in the V2 Mini delivered 2.4× more thrust and 1.5× greater fuel efficiency. **These specs set the performance benchmark our optimized design aims to surpass.**

Motivation

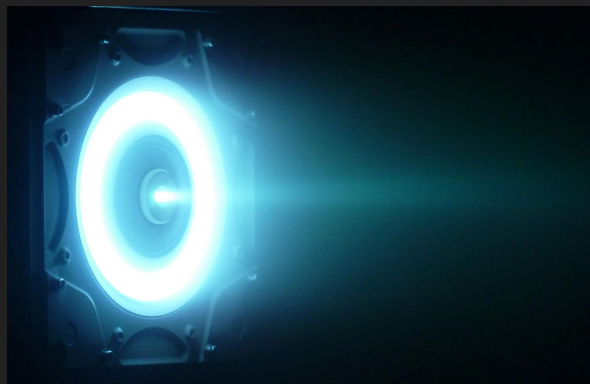
Why Hall Thrusters?

- Higher Isp and low fuel cost, good for smaller satellites
- Fuel efficient, larger payload ratios
- Ideal for longer mission lifetimes



What Drove Our Research?

- Aim to improve efficiency and lower cost
- Improve issues with plume divergence
- Determine alternative types of viable propellant injection



SPACE 

Propellant Discussion

Why Not Argon? The Case for True Alternatives

Argon's Fundamental Limitations

Storage density:

Only 16% of xenon density at same pressure creates a tank volume penalty at constellation scale

Ionization:

Low atomic mass + high ionization energy → short neutral residence time, high flow rates required

The SpaceX claim:

>40% efficiency reported, but no technical papers have been publicly released. Cannot be verified or reproduced.

Our Proposal: A Better Path

SpaceY: Solid Iodine (I₂)

- Similar atomic mass & ionization energy to xenon
- Solid storage (3× denser than pressurized xenon)
- Only 5-10 W heater power to sublimate
- Proven in-orbit (2021)

Future: In-Situ Resources

- Magnesium in lunar & Martian regolith
- Water ice confirmed at lunar south pole
- Eliminates Earth-based supply chain entirely
- Enables permanent deep-space infrastructure

Solid Iodine (The Present Solution)

Why Iodine Works

Atomic Mass

≈ xenon (131.3 amu)

126.9 amu

1st Ionization Energy

lower than xenon (12.1 eV)

10.5 eV

Storage Density

3× pressurized xenon

4.9 g/cm³

Heater Power

to reach operational vapor pressure

5–10 W

Xe

~\$2,300/kg

Ar

~\$900/kg

I₂

~\$30/kg

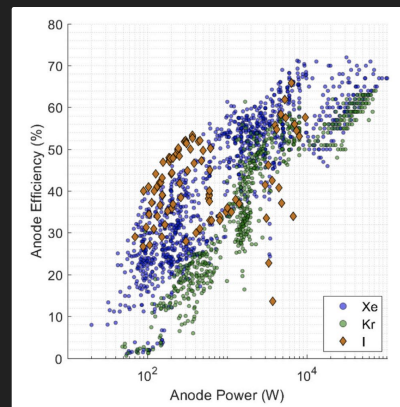
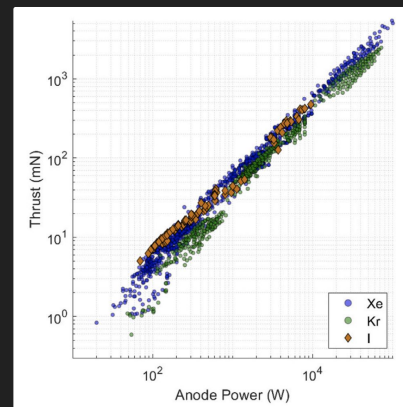
Measured Performance

257 mN

Ideal Thrust

65%

Anode Efficiency



Figures 2 & 3: Thrust (left) and Anode Efficiency (right) vs. Anode Power for Xe and Iodine

Simulated Results for Solid I₂

Xe vs. I ₂ HET Thruster			
Total Power Rating	2.5 kW		
Delta V	2592 m/s		
Dry Mass	1648 kg		
	Xe	Solid I ₂	% Change
Burn Time (h)	4862.9	4941.0	1.6%
Propellant Mass (kg)	239.2	234.9	-1.8%
Propellant Volume (L)	114.4	47.6	-58.3%
Flow Rate (mg/s)	13.7	13.2	-3.3%
Ideal Isp (s)	1950.8	1984.2	1.7%
Ideal Thrust (mN)	231.4	257.0	11.1%
Total Impulse	4574.5	4570.2	-0.1%
Tank Power (W)	0	5	
Propellant Cost (\$)	\$ 549,911.00	\$ 7,515.00	-98.6%

Table 1: Simulated HET Results with Xe and I₂

Future Vision: In-Situ Propellant Utilization

What if we stopped launching propellant from Earth entirely?

Magnesium (Mg)

Moon & Mars

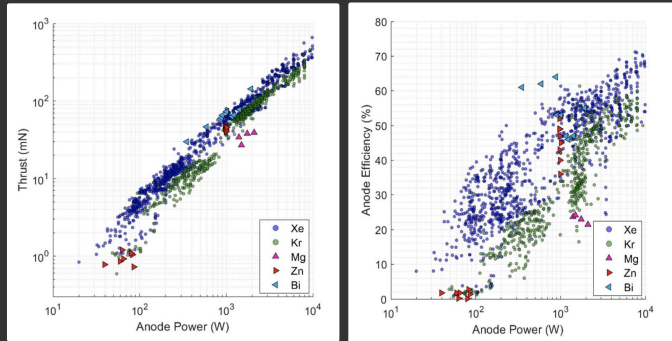


Figure 4: Thrust (mN) and Anode Efficiency (%) in terms of Anode Power (W) for Magnesium (Mg)

- Confirmed in lunar & Martian regolith so no Earth supply needed
- Isp > 2,700 s at 300 V in xenon-optimized thruster
- Anode efficiency 23% in Xe-designed thruster — higher with Mg-optimized geometry
- Stores as non-pressurized solid, similar density to xenon

Water (H₂O)

Lunar Ice & Solar System

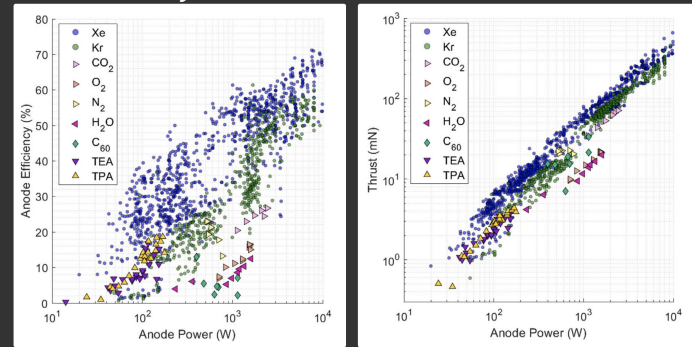


Figure 5: Thrust (mN) and Anode Efficiency (%) in terms of Anode Power (W) for Water (H₂O)

- Confirmed water ice at lunar south pole (accessible in-situ)
- Cathode can be fed by extracted hydrogen, reducing system complexity
- Wide solar system availability (asteroids, comets, outer planet moons)

Divergence Of Plume

Modified Magnetic Topology

What is Magnetic Topology:

- Geometric arrangement and curvature of magnetic field lines caused by the cooperation of magnetic screens with different magnetic pole

Most optimization focuses on:

- Discharge channel B field
- Exit Plane shaping
- Not near-field plume topology

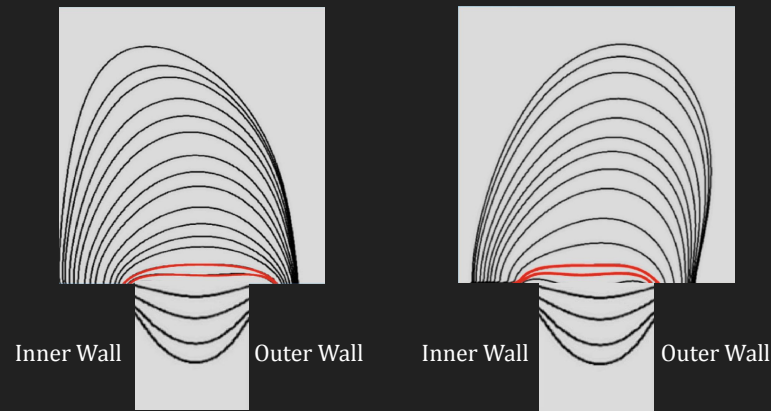


Figure 6: Two types (a) Inclined inward and (b) Inclined outward of magnetic field distributions

$$\alpha_{90} = \arctan \frac{H \sin(S/H) - R_{outer}}{H \cos(S/H)}$$

Defines plume divergence as the angle containing 90% of total ion current leaving the thruster

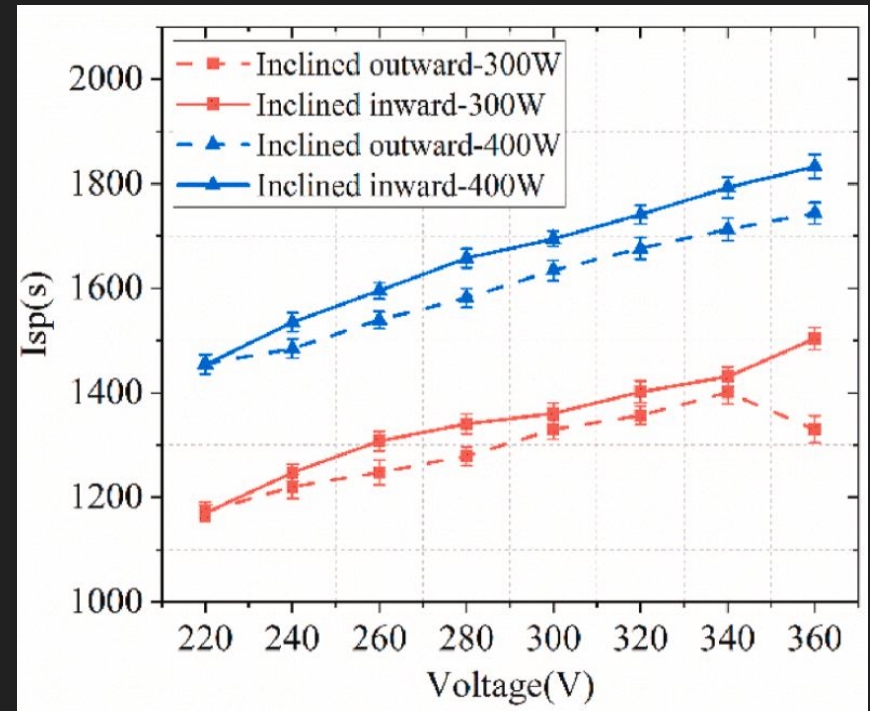
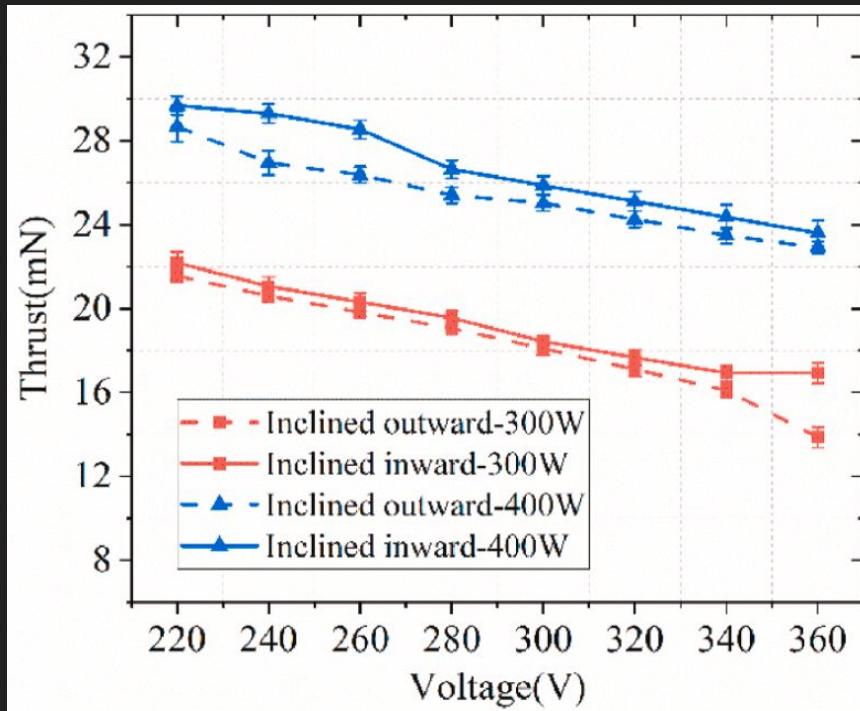


Figure 7: Thrust (left) and Isp (right) vs Voltage at constant power

Under constant power operation (**300 W & 400 W**), the inward-inclined topology **consistently outperformed** the outward configuration, producing **higher thrust, specific impulse, and efficiency** across all discharge voltages tested

Inward-Inclined Reduced Plume

Condition	Thrust (mN) & % Δ	I_{sp} (s) & % Δ	Anode Efficiency (%) & Δ	Plume Half-Angle (°) & Δ
Inclined Inward	18.0 / +3.45%	1410 / +3.30%	53.0 / +6.85%	26.5 / -4.4°
Inclined Outward	17.4 / –	1365 / –	49.6 / –	30.9 / –

Table 2: PIC simulation performance comparison between inward- and outward-inclined magnetic field topologies, outward being the baseline of comparison

Inward inclined topology benefited every major category, regardless of constant flow rate of propellant.

Plume Divergence:

$42.3^\circ \rightarrow 35.3^\circ$

Roughly **7° decrease** (from outward to inward) @ 240 V, 400 W, aligns thrust vector more

Δ Thrust = **+0.6 mN**, Δ I_{sp} = **+45 s**

Post-exit magnetic shaping improved both efficiency AND plume divergence

Propellant Injection

Neutral Injection: Axial vs. Radial

Current: Axial Injection

- Axial/anode-fed injection
- Moderate radial non-uniformity
- Ionization concentrated near centerline
- Neutral atom escape before ionization is possible

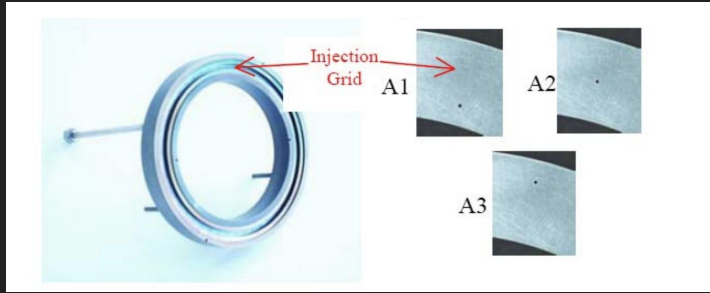


Figure 8: Inner (A1) vs. axial (A2) vs. Outer (A3) Injection

Proposed: Combined Radial Injection

- Injection from inner + outer walls
- Improved radial uniformity
- Better overlap with ionization region
- Increased residence time in active zone

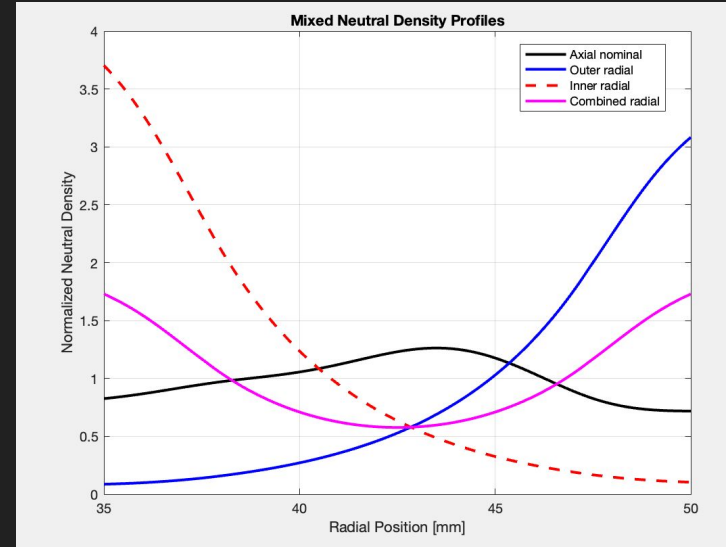


Figure 9: Neutral density profile optimized in combined radial

Key Takeaway: Combined radial injection could produce the most uniform neutral distribution across the channel

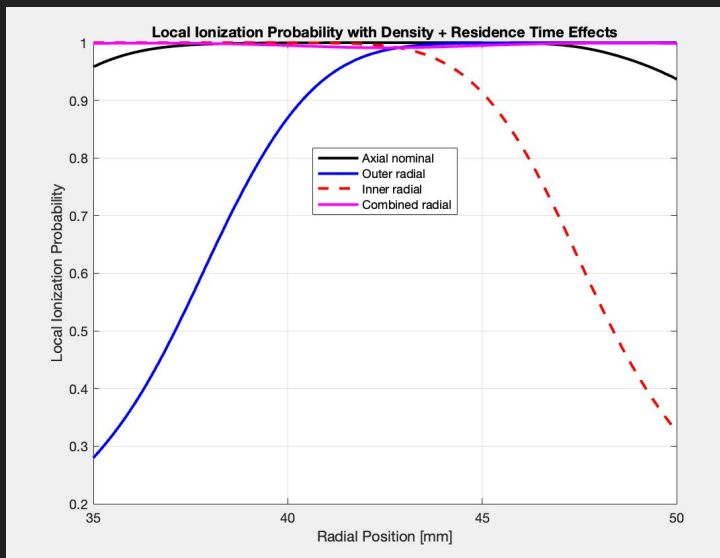


Figure 10: Combined injection lead to more uniform ionization

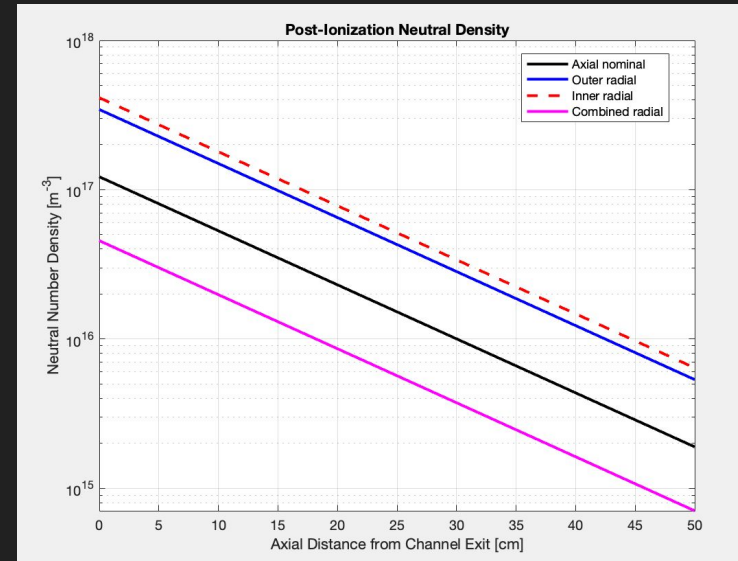


Figure 11: Combined injection lead to lower neutral density

Metric	Axial (Current)	Radial (Single)	Combined Radial (Proposed)
Ionization Uniformity	Moderate	Poor	High
Propellant Utilization	High	Slightly Lower	High (~ same)
Plume Neutral Density	Moderate	High	Low
Plume Divergence	Higher	Higher	Reduced (lit.)

Table 3: Injection scheme comparison across categories

Proposed injection improves plume quality and lifetime without sacrificing efficiency

SpaceY Product Plan

Propellant: Solid Iodine (I₂)

Isp: ~2,000s

Peak Anode Efficiency: 53%

Storage Density: 3× pressurized xenon

Cost: \$32/kg vs. Xe (~\$thousands/kg)

Heritage: Proven in-orbit (2021)

Magnetic Topology

ΔPlume Angle: 42.3° → 35.3° (~ -7°)

Thrust: +0.6 mN (+3.45%)

Isp: +45 s (+3.30%)

Anode Efficiency: +6.85%

Inward topology improves every metric

Combined Radial Injection

Ionization Uniformity: High (vs. Moderate axial)

Plume Neutral Density: Low (vs. Moderate axial)

Plume Divergence: Reduced vs. axial baseline

Propellant Utilization: ~same as axial (High)
Improved lifetime via reduced wall erosion

Open Questions

Iodine × modified geometry interaction: The combined effect of iodine's heavier atomic mass on modified magnetic and injection geometry has not been experimentally validated

Wall Interaction with Radial Injection: Radially injected neutrals will interact with the channel wall in a different manner than current axial injection schemes, next steps would include investigation of surface chemistry effects.

SPACE The logo for SpaceX, featuring the word "SPACE" in a white, sans-serif font, followed by a stylized white rocket ship icon. The rocket ship is depicted in profile, pointing upwards and to the right, with a long, curved line trailing behind it to represent motion or a trajectory.

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Appendix: Propellants 1

Table 1

Physical properties of several previously investigated propellants for Hall thrusters. *Data at 14 MPa, 50 °C, from NIST database. [21,49].

	Xe	Kr	I	Mg	Zn	Bi	Cd	Hg	Cs
Mass (u)	131.3	83.8	126.9	24.3	65.4	209	112.4	200.6	132.9
Ionization Properties									
First ionization energy (eV)	12.1	14	10.5	7.6	9.4	7.3	8.9	10.4	3.9
Peak cross section ($\text{\AA}^2$)	4.8	3.7	6	8	5	8	–	8.3	9.4
Storage and Handling									
Density STP (g/cm ³)	1.6*	0.5*	4.9	1.7	7.1	9.8	8.7	13.5	1.9
Melting point (°C)	–112	–157	113.7	650	420	271	321	–39	28
Vapor pressure at melting point (Pa)	–	–	2.34e+4	384.01	21.49	7.5e–3	15.34	–	–
Toxicity/difficulty to handle	–	–	Med.	Med.	Low	Low	High	High	High

Appendix: Propellants 2

Table 2

Price evolution for selected elements. +Price derived from flask data; * according to U.S. Geological Survey 2022 [98] ** according to Merck ; *** Various independent suppliers;

Purity 99.99%	Market	2017	2018	2019	2020	2021	2022
USD/l							
Xe**	European Market	–	–	–	–	910.97	1,050.00
Kr**	European Market	–	–	–	–	177.86	204.00
Ar**	European Market	–	–	–	–	218.91	243.00
O ₂ **	European Market	–	–	–	–	169.88	195.00
N ₂ **	European Market	–	–	–	–	169.88	195.00
CO ₂ *	European Market	–	–	–	–	182.42	195.00
USD/kg							
Zn*	North American	3.07	3.11	2.74	2.44	3.20	–
	London Metal Exchange	2.89	2.93	2.55	2.26	3.00	–
Cd*	North American	1.75	2.89	2.67	2.29	2.49	–
Mg*	U.S. Spot Western	4.74	4.78	5.40	5.49	8.60	–
	European Free Market	2.27	2.55	2.43	2.15	5.50	–
Bi*	North American	10.89	10.16	7.01	6.00	8.05	–
I ₂ *	North American	19.55	22.46	26.38	31.57	32.00	–
Cs**	North American	–	–	–	65200.00	69900.00	–
	London Metal Exchange	–	–	–	–	–	–
Hg*	European Union	30.20	31.91	–	–	–	–
	Global	36.93	78.59	73.98	–	–	–
USD/kg							
TEA***		–	–	–	–	150.47	398.07
TPA***		–	–	–	–	66.00	113.70
Adamantane***		–	–	–	–	540.00	721.36
Fullerene***		–	–	–	–	493.30	1,000.00

Appendix: Propellants 3

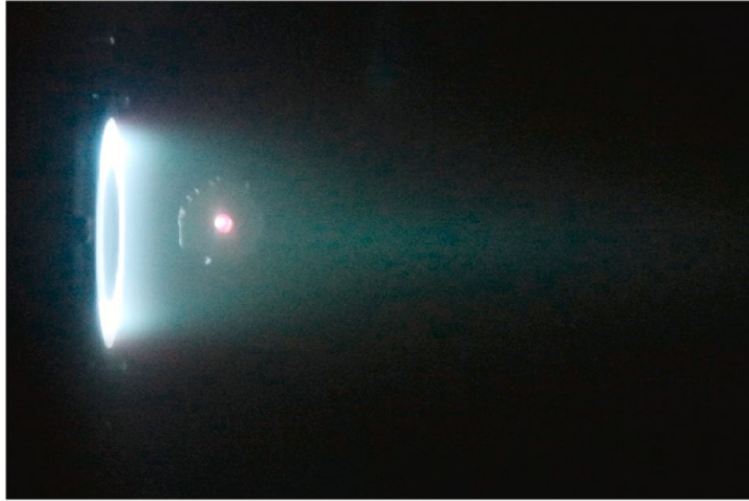
Table 6

Scenario 3 — Psyche mission scenarios for 1 of the 4 thrusters operating at 250 V–2.5 kW (4.5 kW Hall thruster).

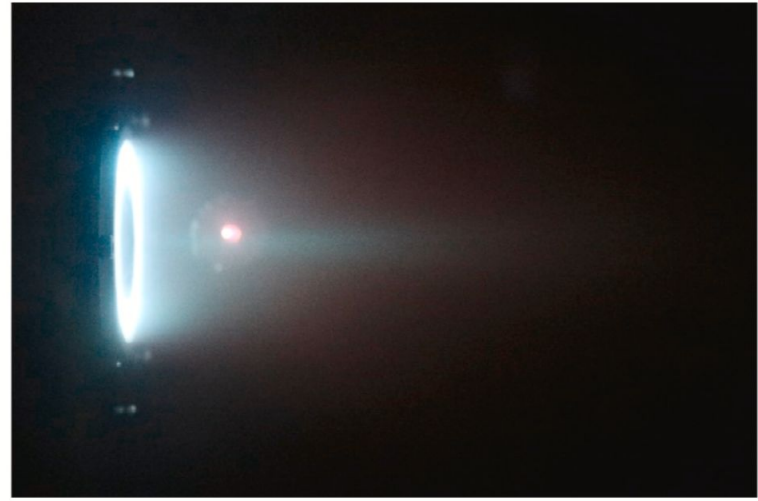
Power Rating (W)	2500						
Voltage (V)	250						
Delta V (m/s)	2592						
Dry Mass (kg)	1648						
	Xe	Kr	I	Mg	Zn	Bi	Cd
Burn Time (h)	4862.85	6002.84	4940.45	10867.43	6752.44	3925.14	5228.31
Propellant Mass (kg)	239.17	188.44	234.87	98.95	165.38	307.28	220.17
Propellant Volume (l)	114.35	208.09	47.64	56.87	23.16	31.33	25.45
Flow Rate (mg/s)	13.66	8.72	13.21	2.53	6.80	21.75	11.70
Ideal Isp (s)	1950.75	2441.77	1984.20	4533.93	2764.39	1546.22	2108.23
Ideal Thrust (mN)	261.36	208.81	256.96	112.45	184.44	329.74	241.84
Total Impulse (kNs)	4575.50	4512.33	4570.16	4399.48	4483.43	4659.44	4551.90
Tank Power (W)	-	-	≈5	401.94	131.34	2981.18	77.08
Propellant Cost (USD)	549911.29	176399.31	7515.77	850.95	529.22	2473.64	548.22

Plume Divergence Appendix

Plume Divergence Visualization



(a) Inclined inward



(b) Inclined outward

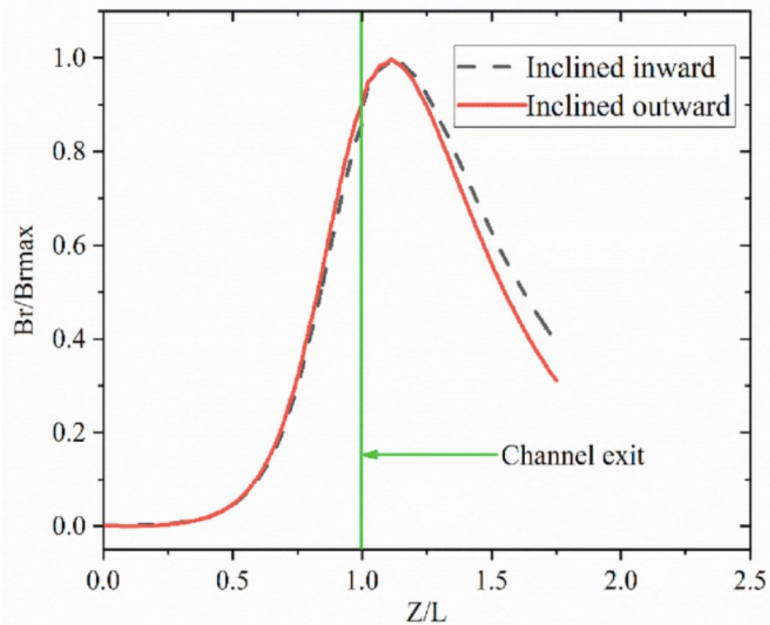
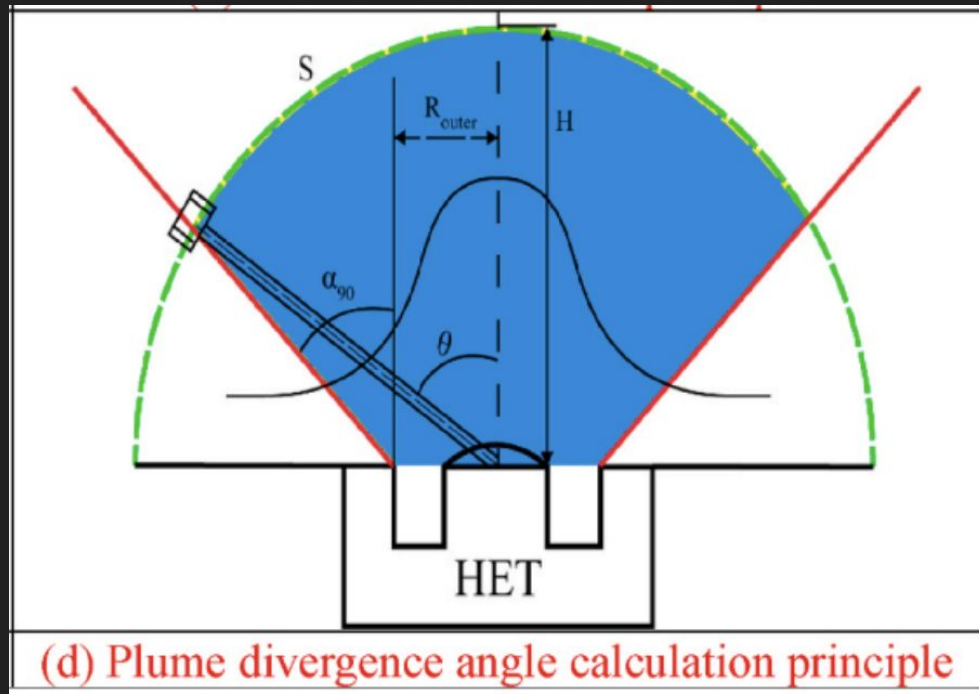
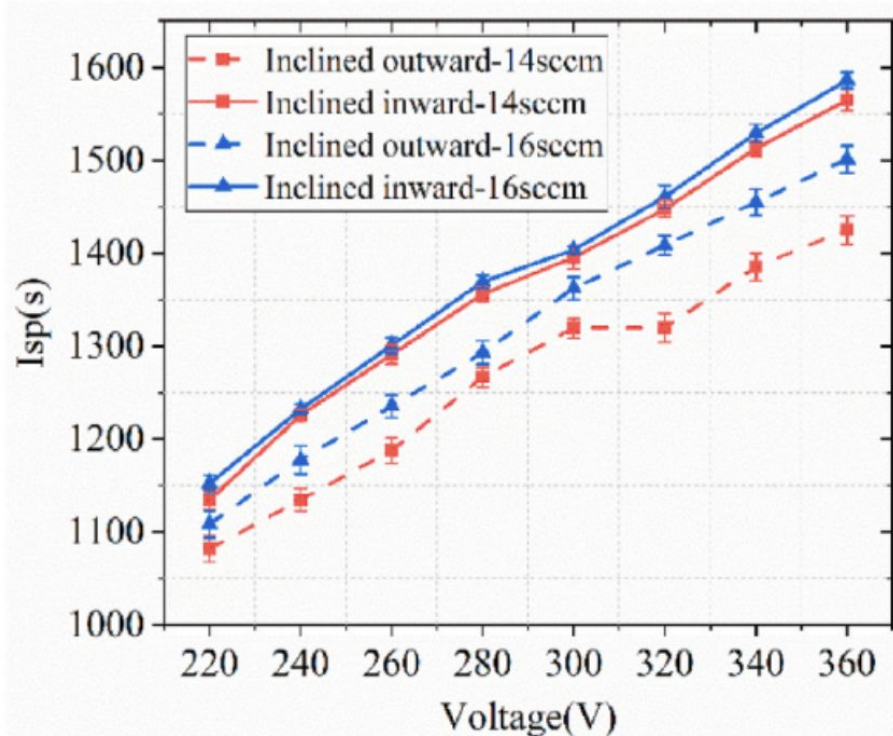
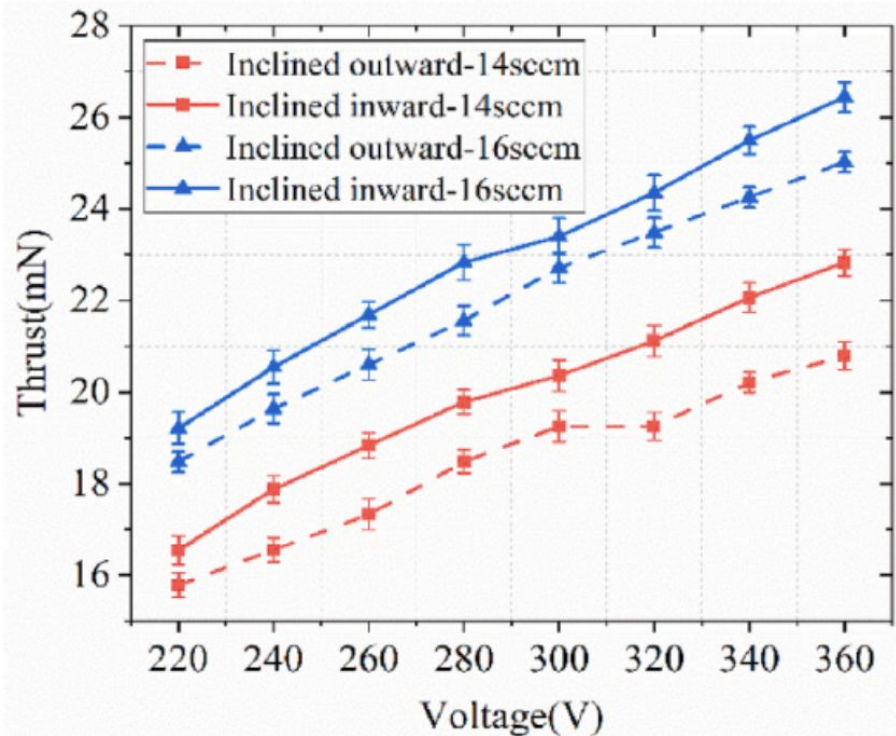


Fig. 2. Magnetic field intensity distribution along channel centerline.



(d) Plume divergence angle calculation principle



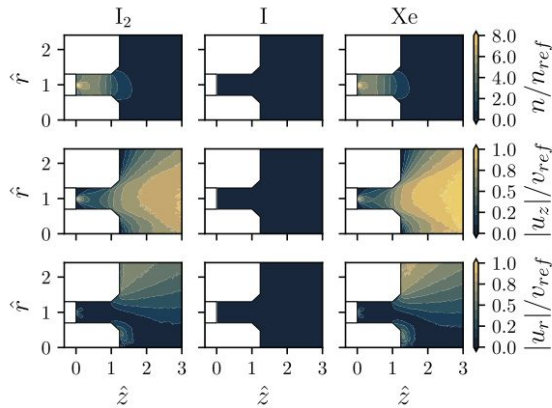


Fig. 5 Conventional channel and axial injection. Maps of (top) particles density, (middle) axial velocity, and (bottom) radial velocity with $T_n = 400$ K. Values for n_{ref} and v_{ref} are computed using Eqs. (9) and (10).

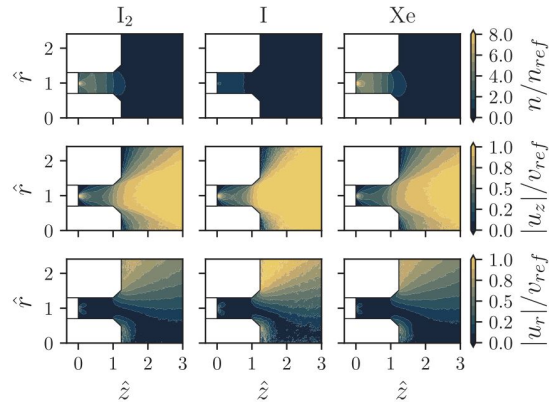


Fig. 6 Conventional channel and axial injection. Maps of (top) particles density, (middle) axial velocity, and (bottom) radial velocity with $T_n = 700$ K. Values for n_{ref} and v_{ref} are computed using Eqs. (9) and (10).

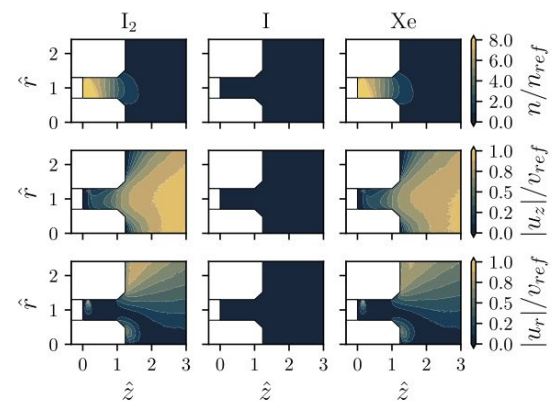


Fig. 9 Conventional channel and radial injection. Maps of (top) particles density, (middle) axial velocity, and (bottom) radial velocity with $T_n = 400$ K. Values for n_{ref} and v_{ref} are computed using Eqs. (9) and (10).

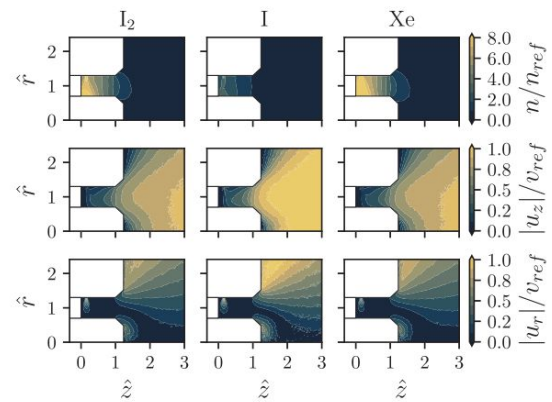


Fig. 10 Conventional channel and radial injection. Maps of (top) particles density, (middle) axial velocity, and (bottom) radial velocity with $T_n = 700$ K. Values for n_{ref} and v_{ref} are computed using Eqs. (9) and (10).